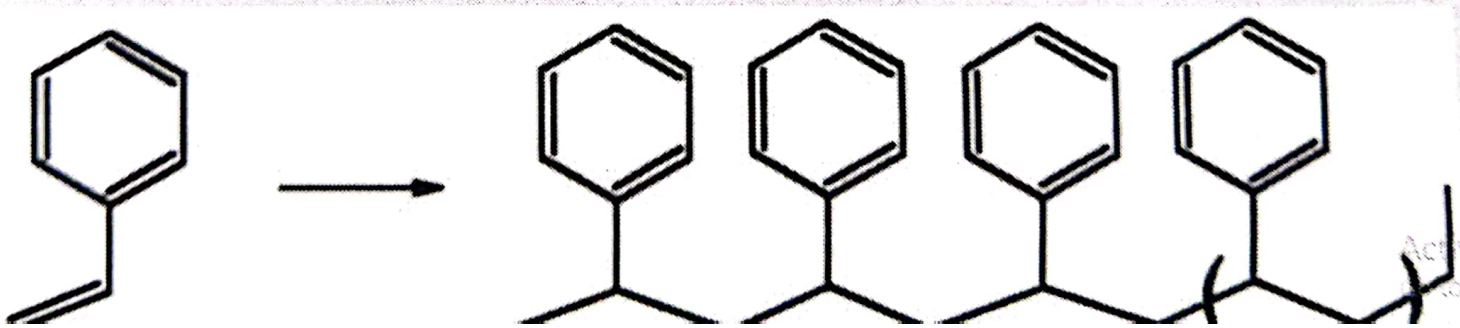
Molecular Weights of Polymers

Molecular weight is one of the most central aspects of polymer properties. Of course, all molecules have molecular weights of their own. It might seem obvious that molecular weight is an essential property of any molecular compound. In polymers, molecular weight takes on added significance. That's because a polymer is a large molecule made up of repeating units

Molecular Weight

- Repetitive units make up a polymer.
- These repetitive units were originally the monomer molecules.
- When polymer chains form their lengths and thus their weights differ.
- It is important to be able to characterize the polymer structure.
- Determining the weight-average molecular weight or the number-average molecular weight is a part of any polymer characterization.



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Number-average molecular weight

- The number average molecular weight is not too difficult to understand.
- It is just the total weight of all the polymer molecules in a sample, divided by the total number of polymer molecules in a sample

$$Mn = \frac{\sum niMi}{\sum ni} = \frac{\sum wi}{\sum wi/Mi}$$

Where,

n = **Moles of molecules ($n_1 + n_2 + n_3 + ------n_i$) i.e. weight (w)/molecular weight (M)**

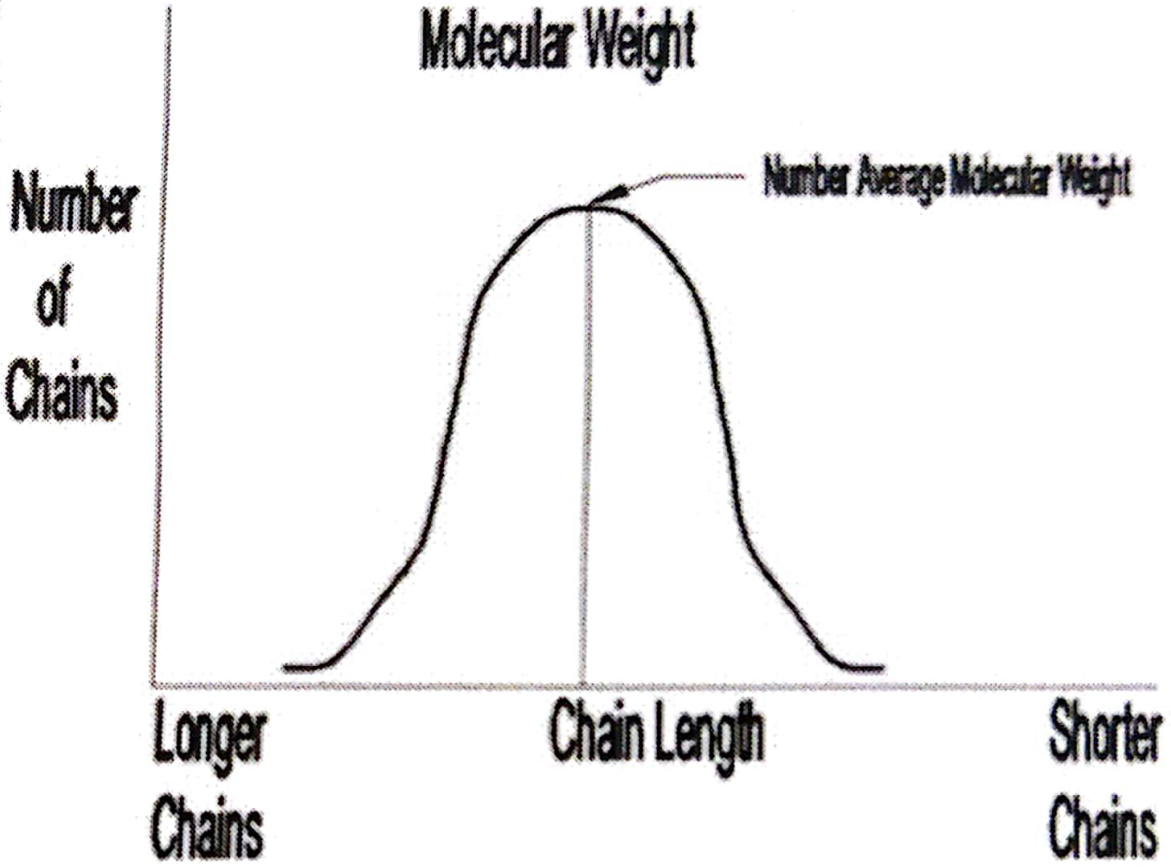
w = **Weight of individual molecules ($w_1 + w_2 + w_3 + -------w_i$)**

M = **Molecular weight of each molecules**

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Molecular Weight

The Number Average Molecular Weight ($\overline{M_n}$) is the total weight of the polymer molecules divided by the total number of polymer molecules.



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Consider a polymer, which contains four molecular weight polymers in different numbers and weight

Polymer entity	Number of unit in each entity, n	Weight of each grams, M(g)	Total weight of each entity, W = nM(g)
Poly-1	2	10	20
Poly-2	4	20	80
Poly-3	6	100	600
Poly-4	3	250	750
Total	15	-	1450

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Total number of polymer in containing each entity of poly-1, poly-2, poly-3 and poly-4 is = 15

Number of Poly-1 present in the polymer = 2

Number of fraction of poly-1 = 2/15

Similarly, Number of fraction of poly-2 = 4/15, Number of fraction of poly-3 = 6/15, Number of fraction of poly-4 = 3/15

Contribution made by poly-1 towards the average weight of polymer = numbe of fraction of each polymer x weight of each poly entity

Therefore, each poly contribution is

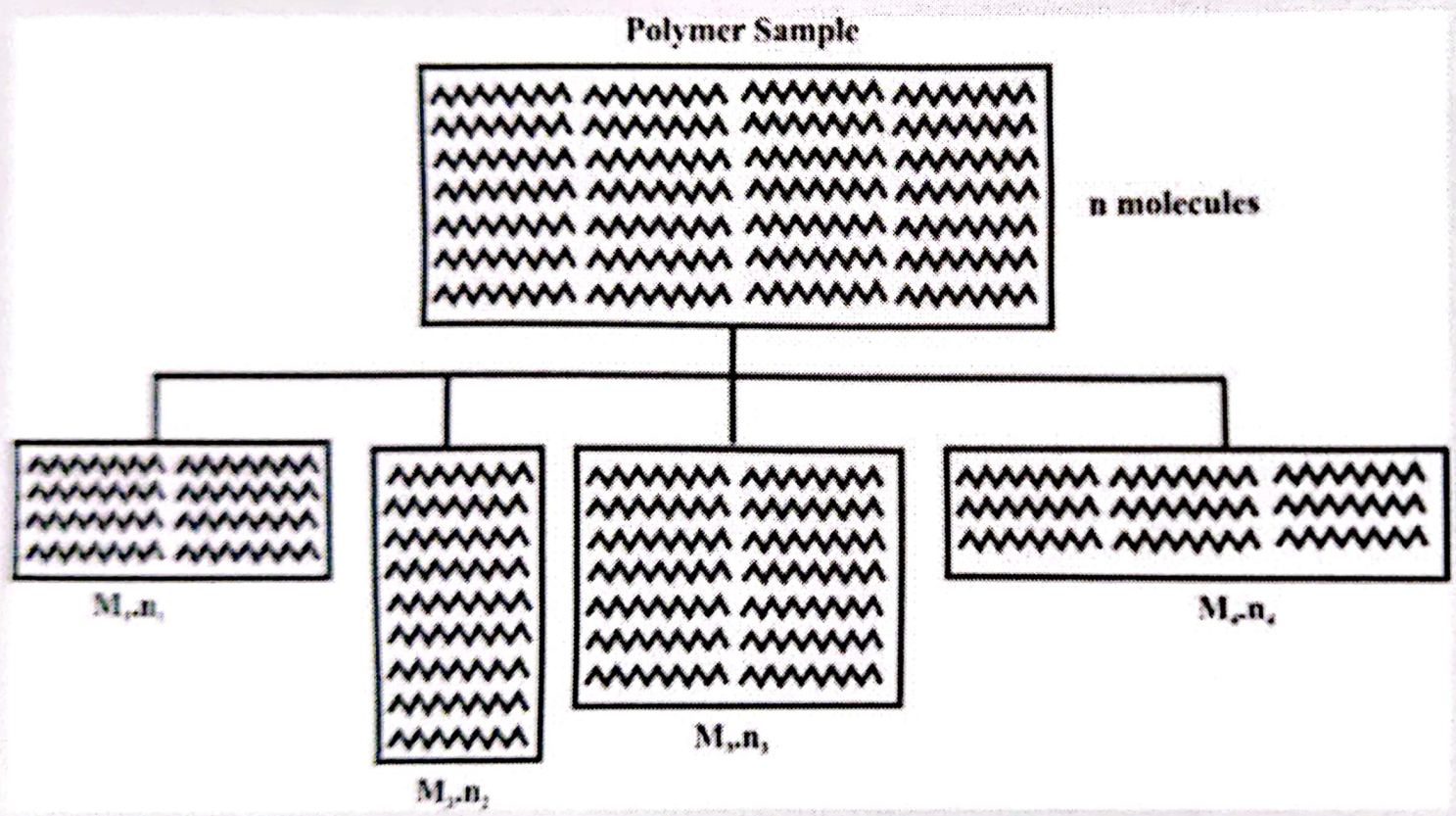
(2/15) x 10 =1.33g, (4/15) x 20 = 5.33g, (6/15) x 100 =40g, (3/15) x 250 = 50

Summing up the contribution to get Number Average Molecular Weight=

1.33 + 5.33 + 40 + 50 = 96.66g

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Molecular weight of a polymer is defined as sum of the atomic weight of each of
e atoms in the molecules, which is present in the polymer".



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Generalization of concept

Total number of molecules (n) id given by

$$n = n_1 + n_2 + n_3 + n_4 + \ldots\ldots\ldots = \sum n_i$$

Number fraction of each molecule is = $\frac{n_i}{\sum n_i}$

Number average weight contribution of each entity is = $\frac{n_i M_i}{\sum n_i}$

Number average weight molecular weight is

$$\frac{n_1 M_2}{\sum n_i} + \frac{n_2 M_2}{\sum n_i} + \frac{n_3 M_3}{\sum n_i} + \frac{n_4 M_4}{\sum n_i} \ldots\ldots = \frac{\sum n_i M_i}{\sum n_i} = M_n$$

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Calculation

$$M_n = \frac{\sum n_i M_i}{\sum n_i}$$

$$M_n = \frac{(2 \times 10) + (4 \times 20) + (6 \times 100) + (3 \times 250)}{2 + 4 + 6 + 3}$$

$$M_n = \frac{20 + 80 + 600 + 750}{15}$$

$$M_n = \frac{1450}{15}$$

$$M_n = 96,67\ g \quad (2\ d.p)$$

Weight-average molecular weight

Total weight of each poly present in the polymer =1450g

Weight of poly-1 present in polymer = 20g

Weight fraction of poly-1 = 20/1450, Weight fraction of poly-2 = 80/1450, Weight fraction of poly-3 = 600/1450, Weight fraction of poly-4 = 750/1450

Contribution made by each poly towards average weight of polymer = weight fraction of poly-1x weigh of each unit

For poly-1 (20/1450) x 10 = 0.14g

For poly-2 (80/1450) x 20 = 1.10g

For poly-3 (600/1450) x 100 = 41.38g

For poly-4 (750/1450) x 250 = 129.31g

Summing up the contribution made by each poly to get weight average molecular weight is

0.14 + 1.10 + 41.38 + 129.31 = 171.93g

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Generalization of concept

- Total number of molecules (n) is given by

$$n = n_1 + n_2 + n_3 + n_4 + \ldots\ldots\ldots = \sum n_i$$

Total weight of the polymer is = $\sum N_i M_i$ W

- Weight fraction of each molecule is = $\frac{n_1 M_1}{W} = \frac{n_1 M_1}{\sum n_1 M_1}$

- Weight average weight contribution of each entity is = $\frac{n_1 M_1 M_1}{\sum n_i M_i} \frac{n_i M_i^2}{\sum n_i M_i}$

- Number average weight molecular weight is

$$\frac{n_1 M_1^2}{\sum n_i M_i} + \frac{n_2 M_2^2}{\sum n_i M_i} + \frac{n_3 M_3^2}{\sum n_i M_i} + \frac{n_4 M_4^2}{\sum n_i M_i} \ldots\ldots = \frac{\sum n_i M_i^2}{\sum n_i M_i} = M_w$$

- For synthetic polymers Mw is greater than the Mn. If they are equal than they will consider as perfectly homogeneous. (Each molecule has same molecular weight).

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$$M_w = \frac{\sum n_i M_i^2}{\sum n_i M_i}$$

$$= \frac{(2 \times 10^2) + (4 \times 20^2) + (6 \times 100^2) + (3 \times 250^2)}{(2 \times 10) + (4 \times 20) + (6 \times 100) + (3 \times 250)}$$

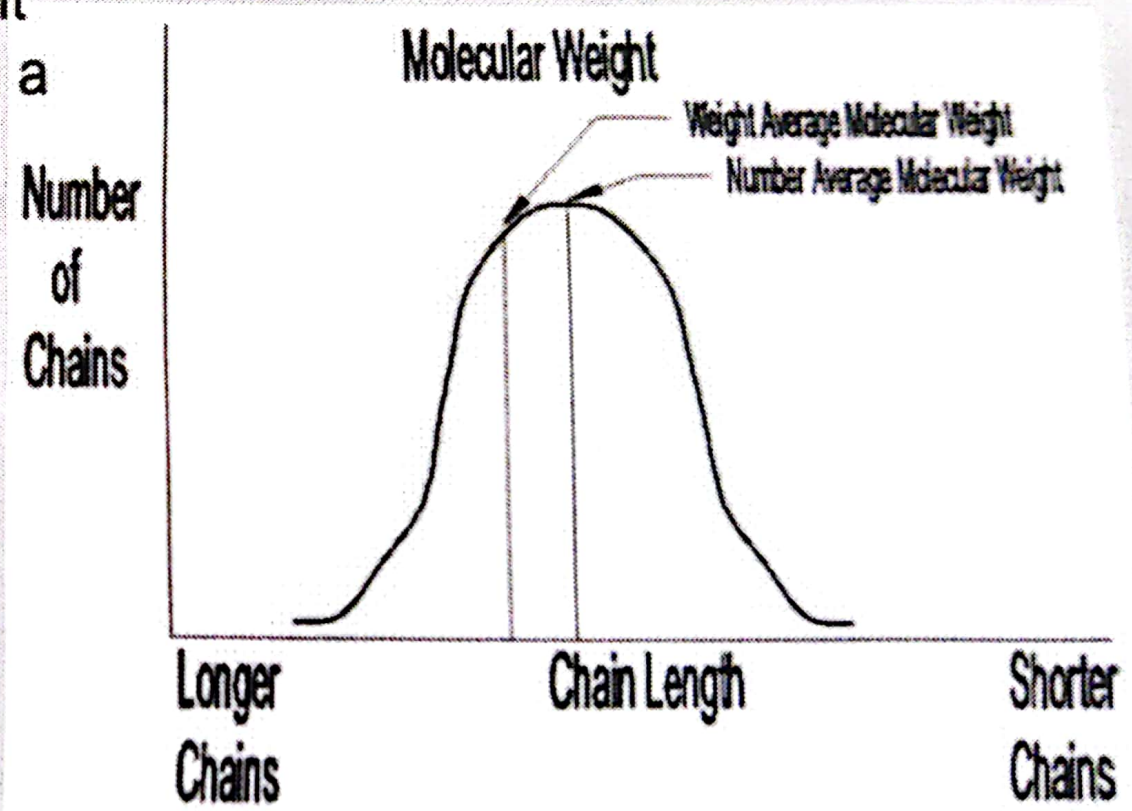
$$= \frac{249\,300}{1450}$$

$$= 171,93 \text{ g.}$$

Molecular Weight

The Weight Average Molecular Weight ($\overline{M_w}$) takes into account that the larger molecules contain a much higher amount of the molecular mass of the polymer.

The Weight Average Molecular Weight is almost always higher than the Number Average Molecular Weight ($\overline{M_n}$).

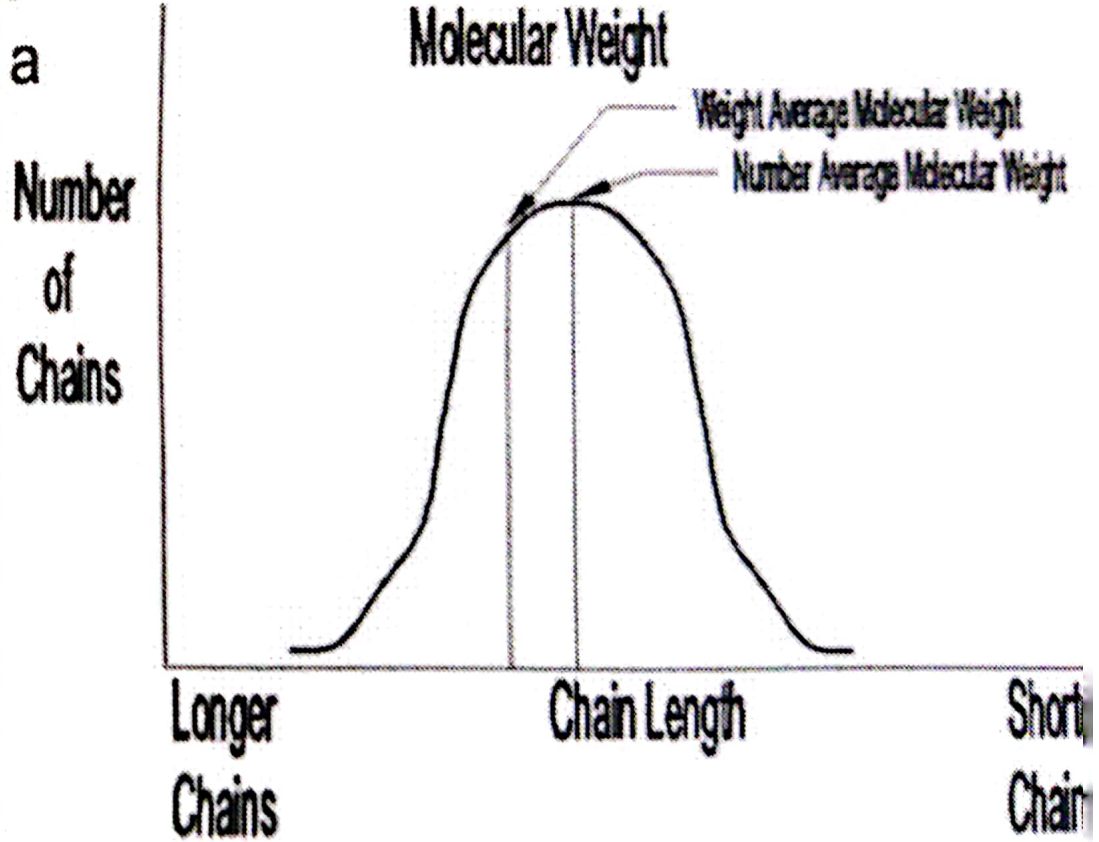


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Molecular Weight

The Weight Average Molecular Weight ($\overline{M_w}$) takes into account that the larger molecules contain a much higher amount of the molecular mass of the polymer.

The Weight Average Molecular Weight is almost always higher than the Number Average Molecular Weight ($\overline{M_n}$).



3.3 SEDIMENTATION AND VISCOSITY-AVERAGE MOLECULAR WEIGHTS

Apart from $\overline{M}_n$ and $\overline{M}_w$, there are two other ways of expressing the molecular weight based on the sedimentation and flow behaviour of the polymer in solution. They are: *z*-average molecular weight ($\overline{M}_z$) and viscosity-average molecular weight ($\overline{M}_v$), expressed as

$$\overline{M}_z = \frac{\Sigma n_i M_i^3}{\Sigma n_i M_i^2} \tag{3.3}$$

and

$$\overline{M}_v = \left(\frac{\Sigma n_i M_i^{1+a}}{\Sigma n_i M_i}\right)^{1/a} \tag{3.4}$$

In these two equations n_i is, as before, the number of molecule having the molecular weight of M_i present in the sample and a i a variable (its value ranging from 0.5 to 1) in the Mark-Houwin equation which relates intrinsic viscosity $[\eta]$ with the viscosity ave rage molecular weight ($\overline{M}_v$). (see Section 12.6.)

We will deal with experimental techniques used for determinin molecular weights in Chapter 13. It will suffice here to know tha light scattering, osmotic pressure, viscosity and ultracentrifugatio methods are used to get $\overline{M}_w$, $\overline{M}_n$, $\overline{M}_v$, and $\overline{M}_z$, respectively.

Influence of Molecular weight of Polymers

The influence of molecular weight on the bulk properties of polyolefins, an increase in the molecular weight leads to

- Increase in:
  - Melt viscosity
  - Impact strength
- Lowers in:
  - Hardness
  - Stiffness
  - Softening point
  - Brittle point

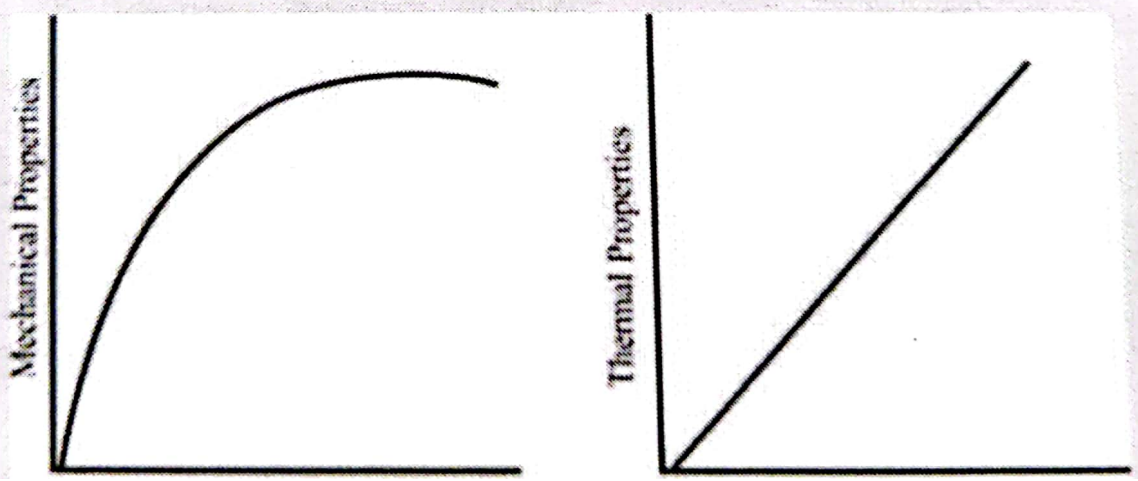
High molecular weight polymer does not crystallize so easily as lower molecular weight material crystallizes due to chain entanglement and that reflect in bulk properties of the high molecular weight polymer.

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•A high molecular weight polymer increases the mechanical properties. Higher molecular weight implies longer polymer chains and a longer polymer chain implies more entanglement thereby they resist sliding over each other.

• Increasing the molecular weight and the chain length of the polymer increases impact strength.

• Thermal properties can also improved by increasing the molecular weight.



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Important Facts

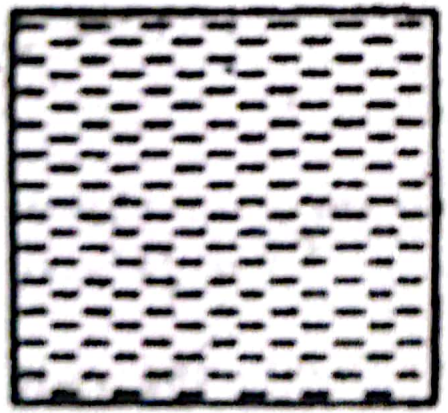
- Weight- average molecular weight is larger or equal to number-average molecular weight.
- Weight- average molecular weight and molecular weight distributions are determined from ultracentrifuge sedimentation, diffusion and light scattering.
- Number-average molecular weight and molecular weight distributions are determined from osmotic pressure and intrinsic viscosity determinations.
- Optical properties are best reflected in the weight-average molecular weight, while strength properties are best reflected in number-average molecular weight.

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3.5 POLYDISPERSITY AND MOLECULAR WEIGHT DISTRIBUTION IN POLYMERS

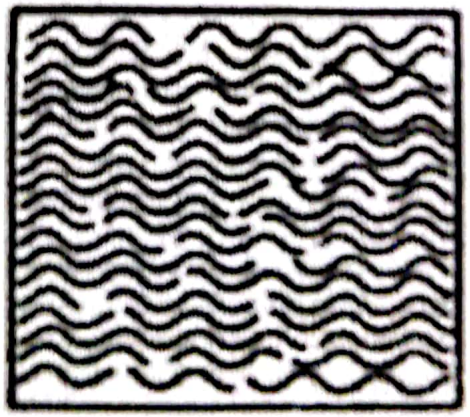
It may oe recalled that a simple chemical compound contains molecules, each of which has the same molecular weight (a 'monodispersed' system), whereas a polymer contains molecules, each of which can have different molecular weights (a 'polydispersed' system). Monodispersity and polydispersity can be represented as shown in Fig. 3.2.

Monodispersed system



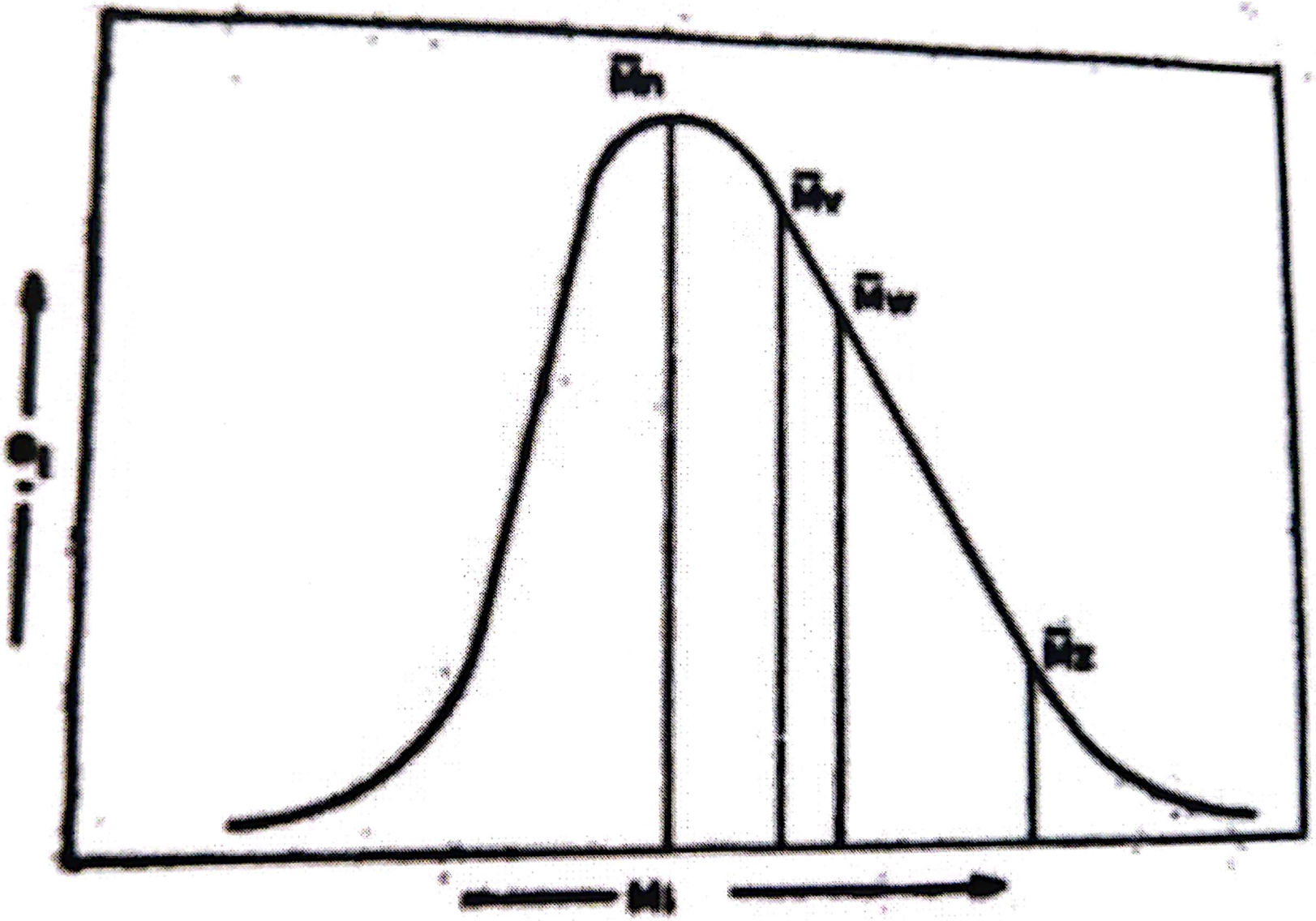
All molecules are of same size

Polydispersed system



Molecules are of different sizes

While the polydisperse nature of the polymer is the basis of the concept of 'average' molecular weight, this 'average' by itself conveys nothing on the dispersity pattern in a given polymer sample. Take a polymer sample of, say, 40,000 average molecular weight. Now, this may mean that the molecules have molecular weights ranging from 20,000 to 80,000 or from 500 to 100,000—the figure is uncertain. This is the reason why two polymer samples of the same 40,000 $\bar{M}_n$ can display similar properties in some respects but not in some others. To know a polymer properly, we must have a knowledge of both the average molecular weight as well as its dispersion pattern. This dispersity with respect to the lowest to the highest molecular weight homologues is expressed by a simple molecular weight distribution curve. Such a curve for a polymer sample is computed by plotting the number fraction (n_i) of molecules having a particular molecular weight (M_i) against the corresponding molecular weight (Fig. 3.3). While the details of computing a molecular weight distribution curve are given in Chapter 13,



we may observe here that such a curve reveals the pattern of the different molecular species present in a polymer sample. The curve shown in Fig. 3.3 brings out the relative positions of various average molecular weights in a polydispersed polymer sample. It may be noted that $\overline{M}_w$ is greater than $\overline{M}_n$ and that $\overline{M}_v$ is closer to $\overline{M}_w$ than to $\overline{M}_n$. Figure 3.4 shows molecular weight distribution in two

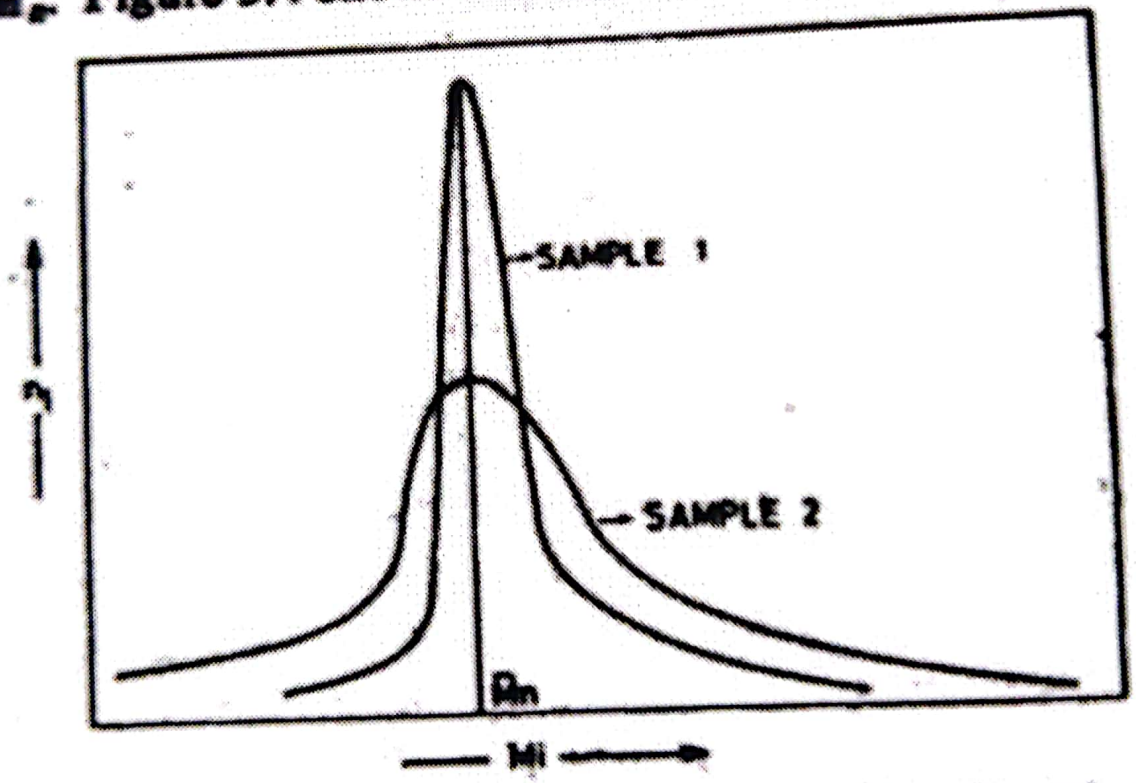


Fig. 3.4 Molecular weight distribution curves of two hypothetical polymer samples with the same number-average molecu-

Polydispersity is a very important parameter. It gives an idea of the lowest and the highest molecular weight species as well as the distribution pattern of the intermediate molecular weight species. To quickly ascertain the degree of polydispersity, we can determine $\overline{M}_w$ and $\overline{M}_n$ by two different experimental methods and then compute $\overline{M}_w/\overline{M}_n$. This ratio is indicative of the extent of polydispersity. For all synthetic polymers, this ratio is higher than 1. As the molecular weight distribution becomes broader (that is to say, as the dispersity increases), the value of $\overline{M}_w/\overline{M}_n$ increases. The three curves in Fig. 3.5 show the dispersity ratio increasing for the three samples with the same $\overline{M}_n$, but a different molecular weight distribution.

Polydispersity index(PDI): Index of polydispersity or PDI is used as a measure of molecular weight distribution and is defined as-

$$PDI = \frac{Mw}{Mn}$$

In case of monodisperse system(natural polymers and synthetic polymers made by anionic polymerization), PDI= 1, Since, Mn= Mw; and for other cases, PDI>1 or Mw is used as a measure of molecular weight distribution and is defined as- Mw>Mn.

Figure in this slide shows molecular weight distribution in two samples having same number average molecular weight but different polydispersities. Sample 1 obviously has narrower dispersion pattern and hence a lower polydispersity than sample 2..

Polydispersity gives an idea of the lowest and highest molecular weight species as well as the distribution pattern of the intermediate molecular species.

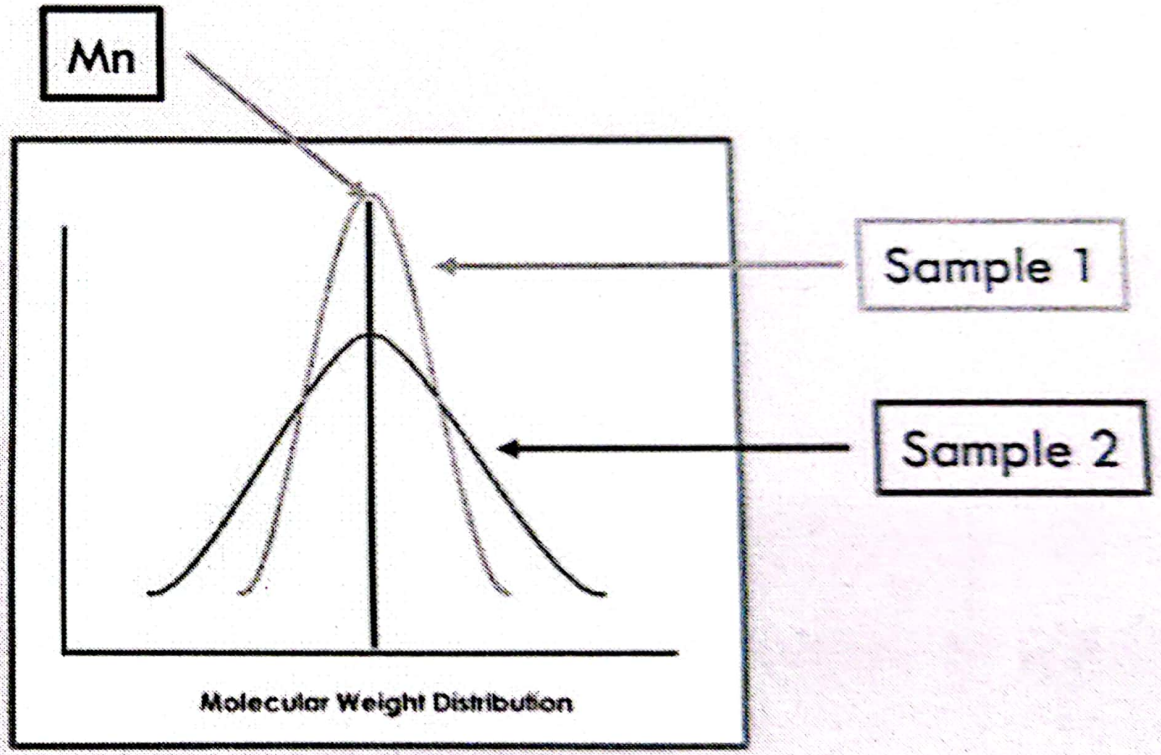
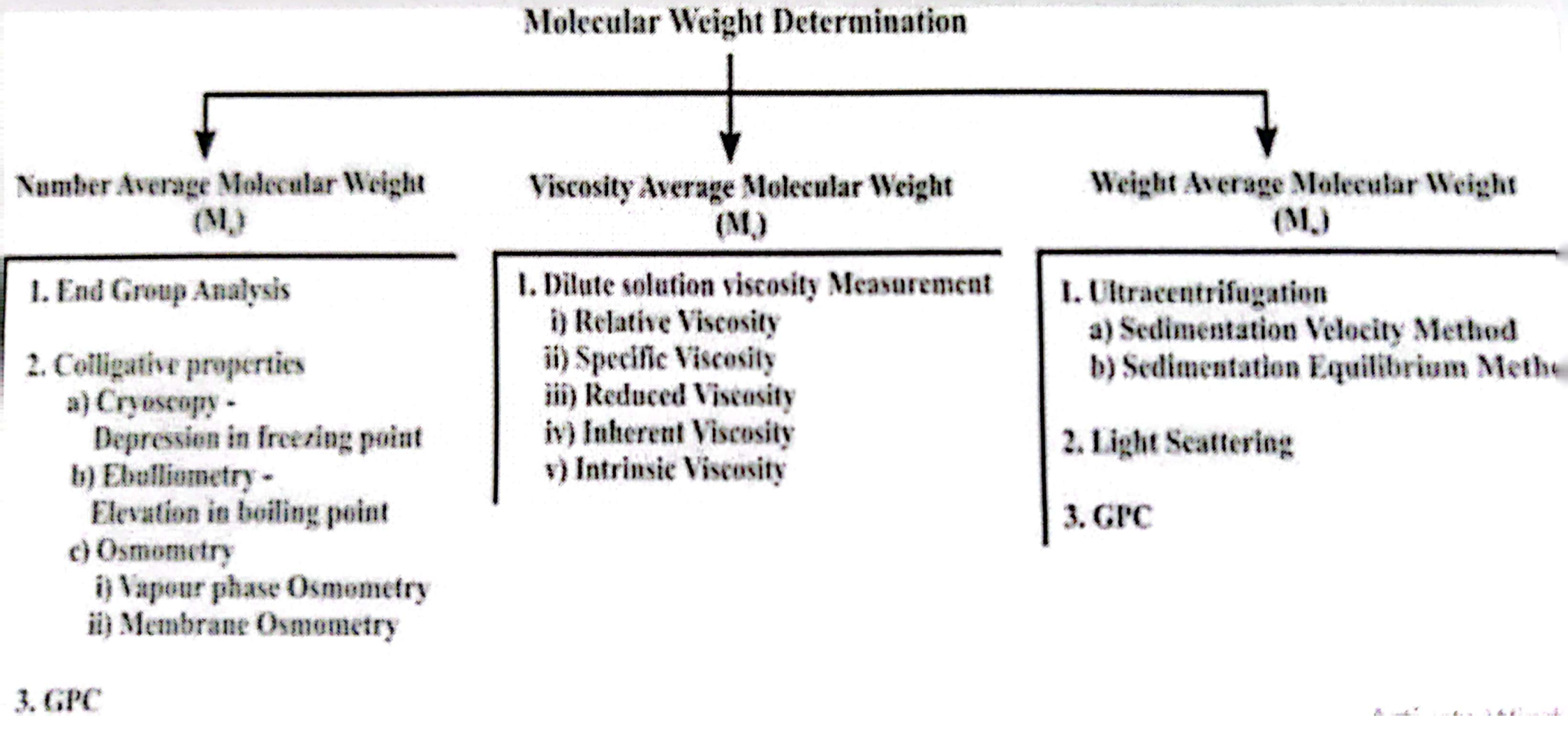


Figure: Molecular weight distribution of two hypothetical polymer samples having same number average molecular weight but different polydispersities.

Determination of Molecular Weight



Methods: End group analysis, viscometry, light scattering and osmotic pressure

End Group Analysis: The end group analysis is a chemical method used to determine the number average molecular weight of a polymer sample through the analysis of reactive functional groups at one end or both end e.g. Carboxyl terminated polybutadiene, hydroxyl terminated polybutadiene. Apart from hydroxyl and carboxyl groups, other end groups (aldehyde, amino, ester) can possibly be present in various polymers. For detecting and analyzing quantitatively functional end-groups of linear polymer (nylon), the amino end-groups of nylon dissolved in m-cresol are readily determined by titration with a methanolic perchloric acid solution. Other titratable end groups are the hydroxyl and carboxyl groups in polyesters and the epoxy end groups in epoxy resins. The sensitivity of this method decreases as the molecular weight increases. This method is limited to the determination of polymers with a molecular weight of less than about 20,000.

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weight of less than about 20,000.

From the knowledge of functional group equivalent and functionality, the molecular weight can be calculated using the following equation-

$$\mathbf{Mn} = \frac{Functionality}{Functional\ group\ equivalent}$$

End Group Analysis: Example: The carboxyl value of cabroxyl terminated polybutadiene (CTPB) can be determined by dissolving a known amount of the polymer in a mixture of toluene and ethanol (1:3) and titrating against alcoholic KOH using phenolphthalein indicator.

Let, the volume of KOH consumed= V ml; Normality of KOH= N; and Weight of the sample taken= w g

$$\text{Carboxyl value} = \frac{V\ x\ N\ x\ 56.1}{w}$$

$$\text{Carboxyl equivalent/100g} = \frac{V\ N\ (56.1) \times\ 100}{w\ (1000) \times\ 56.1}$$

In a particular measurement, if 0.8734 g of the sample has consumed 5.1 ml 0f 0.1242 (N) alcoholic KOH solution then

$$\text{Carboxyl equivalent/100g} = \frac{5.1 \times 0.1242}{10 \times 0.8734} = 0.0725$$

Assuming functionality of 2,

$$\text{Number average Molecular weight; Mn} = \frac{2 \times 100}{0.0725} = 2760$$

The chemical analysis method is limited to the insolubility of the polymer in the solvent

Viscometry: The molecular weight obtained by this technique is the viscosity average molecular weight, Mv.. The viscosity of a polymer solution is considerable high as compared to that of pure solvent. The increase in viscosity by the macromolecules in a solution is a direct function of the hydrodynamic volume and hence the molecular weight of the macromolecules.

The relationship between viscosity of a polymer solution and molecular weight is given by Mark-Houwink equation-

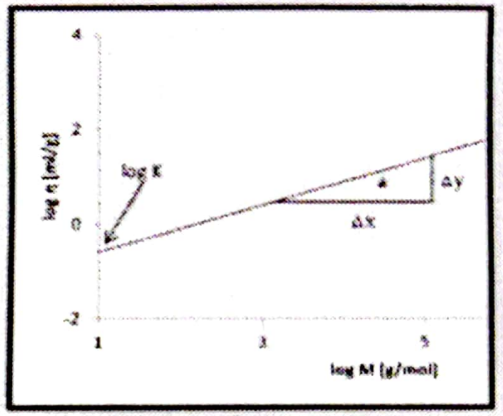
$$[\eta] = KM^a$$

Where [η] is the intrinsic viscosity, M- molecular weight and a and K are constant for a particular polymer/solvent/temperature system. Values of K and a are available for may known polymers.

For a polymer type of known K and a values what is required is the determination of the intrinsic viscosity using the above equation.

For unknown polymer systems, the K and a values are generated by fractionating the polymer sample into several fractions and for each fraction the molecular weight is determined by Osmometry or light scattering method and corresponding intrinsic viscosity is measured

Viscometry: A plot of log [η] against log M gives a straight line. From the graph, the value of K and a can be determined from their ordinate intercept and slope of the line.



$$[\eta] = KM^a$$

$$\log [\eta] = \log K + a \log M$$

Now, let us see how the intrinsic viscosity can be measures. Assume that a liquid is flowing through a capillary tube. The time required for the liquid of volume V to pass through the capillary of radius r and length l is related to its absolute viscosity by the Poiseuille Equation-

$$\eta = \frac{3.14\, Pr^4 t}{8\, V\, l} \Rightarrow t = \frac{8Vl\eta}{3.14Pr^4}$$

Where P is the pressure head under which the liquid flow takes place.

If η and η_o are the absolute viscosities of a solution and the pure solvent respectively and t and to are their corresponding time flow, then ⟶

$$\eta_r = \frac{t}{t_o} = \frac{\eta}{\eta_o}$$

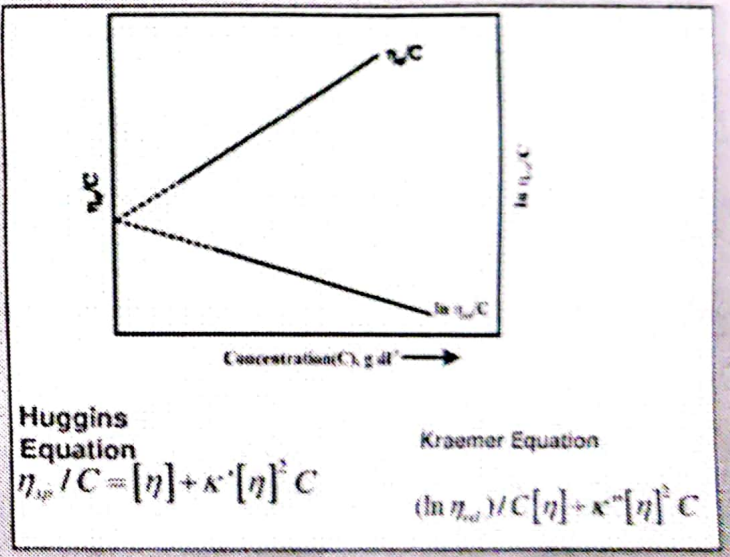
ınd to are their corresponding time flow, then →

η/η_o is known as the relative viscosity, η_r or η_{rel}

Commonly used terms in viscometry:

Relative viscosity	Viscosity ratio	η/η_0	$\eta_{rel} = \eta_r$
Specific viscosity	—	$(\eta/\eta_0) - 1$ or $(\eta - \eta_0)/\eta_0$	η_{sp}
Reduced viscosity	Viscosity number	η_{sp}/c	η_{red} or η_{sp}/c
Inherent viscosity	Logarithmic viscosity number	$\ln(\eta_r)/c$	η_{inh} or $\ln(\eta_r)/c$
Intrinsic viscosity	Limiting viscosity number	$\lim(\eta_{sp}/c)_{c\to 0}$ or $\lim(\ln \eta_r/c)_{c\to 0}$	$[\eta]$ or LVN

Intrinsic viscosity is also known as Staudinger index or limiting viscosity index (dimension is reciprocal of concentration). For calculating the intrinsic viscosity of a polymer sample in solution , we need not know the absolute viscosities of solvent and solution, but only the flow time of constant volume of solvent and the solution through a particular tube. This principle is used in the viscometric technique of molecular weight determination. The term [η] has related to the two viscosity functions through he following two equations by Huggins and Kraemer equations.



Huggins Equation

$$\eta_{sp}/C = [\eta] + k'[\eta]^2 C$$

Kraemer Equation

$$(\ln \eta_{rel})/C = [\eta] + k''[\eta]^2 C$$

K" & k" are constants for a given polymer solvent/ temperature

Determination of Molecular Weight using Colligative Properties

- The use of colligative properties of solution in the determination of the molecular weight of various compounds has been one of the simplest methods. Since colligative properties of solutions depend only on the number of molecules of solute in them, this method is especially used for determining the molar masses of complex molecules, proteins, macromolecules, polymers.
- A solution of known concentration of the substance is prepared in a solvent whose boiling point or freezing point or vapour pressure is known to us. The property to be measured is chosen in such a way that its measurement is simple and easy under the given conditions and do not vary considerably

Elevation of Boiling Point

- We know that the boiling point of a pure solvent rises when a non-volatile solute is added to it. This elevation in boiling point is proportional to the molal concentration of the solute in the solution. Mathematically it is given as:
- $\Delta T_b = K_b m$
- Where,
- ΔT_b = elevation in boiling point
  K_b = *Boiling Point Elevation Constant*
- m = *molal concentration of the solution*

- Now we know that molality is the number of moles of solute per kilogram of the solvent. Let the weight of solute be w_2 having molar mass M_2 dissolved in w_1 grams of solvent. Then molality can be given as:
- Then molality can be given as: m=moles of solute/mass of solvent in Kg
- $= w_2/M_2 / w_1/1000$
- $\Rightarrow m=1000\times w_2/w_1\times M_2$
- So now, elevation of boiling point is given by:
-
- $:M_2 = K_b \times 1000 \times w_2 \ / \ w_1 \times \Delta T_b$
- The above equation gives us a relation between molecular weight and the colligative property that is elevation in boiling point.

Depression of Freezing Point

- The depression in freezing point is another colligative property of solutions that gives us a method to determine the molar masses of various substances similar to elevation of boiling point. We know that depression in freezing point is given as:
- $\Delta T_b = K_b m$
- Where,
- ΔT_f = depression of freezing point
  K_b = Freezing Point Constant
  m = molal concentration of the solution
- From the above description we now know that molality is given as:
- $m = 1000 \times w_2 / w_1 \times M_2$
- Therefore,

- Freezing point depression can be written as:
- $\Delta T_f = K_f \times 1000 \times w_2/w_1 \times M_2$
- Now if we rearrange the above equation, we get:
- $\Rightarrow M_2 = K_f \times 1000 \times w_2/w_1 \times \Delta T_f$
- Thus we can calculate the molar mass of a solute as per the above equation if we can measure the depression in freezing point of the solvent on addition of the solute

Osmotic pressure Method:

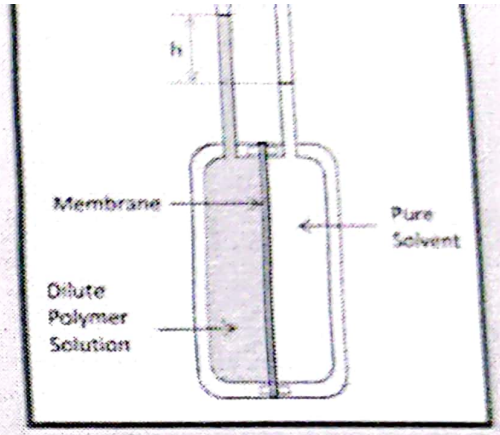
Osmometry: Membrane osmometry is absolute technique to determine M_n. The solvent is separated from the polymer solution with semipermeable membrane that is strongly held between the two chambers. One chamber is sealed by a valve with a transducer attached to a thin stainless steel diaphragm which permits the measurement of pressure in the chamber continuously. Membrane osmometry is useful to determine M_n about 20,000-30,000 g/mol and less than 500,000 g/mol. When M_n of polymer sample more than 500,000 g/mol, the osmotic pressure of polymer solution becomes very small to measure absolute number average of molecular weight. In this technique, there are problems with membrane leakage and symmetry. The advantages of this technique is that it doesn't require calibration and it gives an

advantages of this technique is that it doesn't require calibration and it gives an absolute value of M_n for polymer samples.

Since osmotic pressure is dependent on colligative properties, i.e., the number of particles present, the measurement of this pressure (osmometry) may be applied to the determination of the osmotic pressure of solvents vs. polymer solutions.

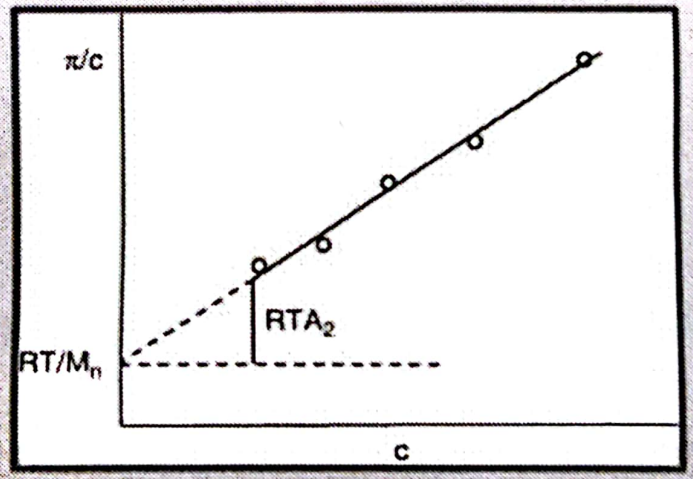
$$\pi = \frac{RT}{M} C + BC^2$$

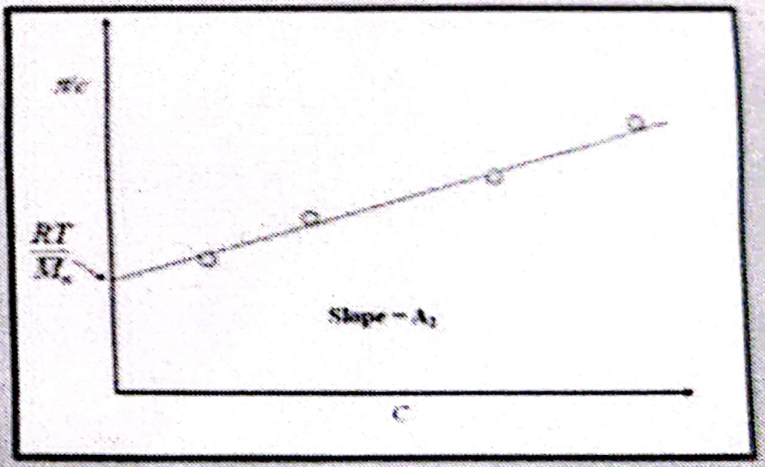
The difference in height (h) of the liquids in the columns may be converted to osmotic pressure (π) by multiplying the gravity (g) and the density of the solution (p), i.e., $\pi = hpg$.



In an automatic membrane osmometer, the unrestricted capillary rise in a dilute solution is measured in accordance with the modified van't Hoff equation:

$$\frac{\pi}{RTC} = \frac{1}{M_n} + A_2 C$$





The reciprocal of the number average molecular weight (M_n) is the intercept when data for π /RTC vs. C are extrapolated to zero concentration.

Light Scattering Method: Light scattering methods to determination of weight average molecular weight, Mw. When polarizable particles are placed in the oscillating electric field of a beam of light, the light scattering occurs. Light scattering method depends on the light, when the light is passing through polymer solution, it is measure by loses energy because of absorption, conversion to heat and scattering. The intensity of scattered light relies on the concentration, size and polarizability that is proportionality constant which depends on the molecular weight..

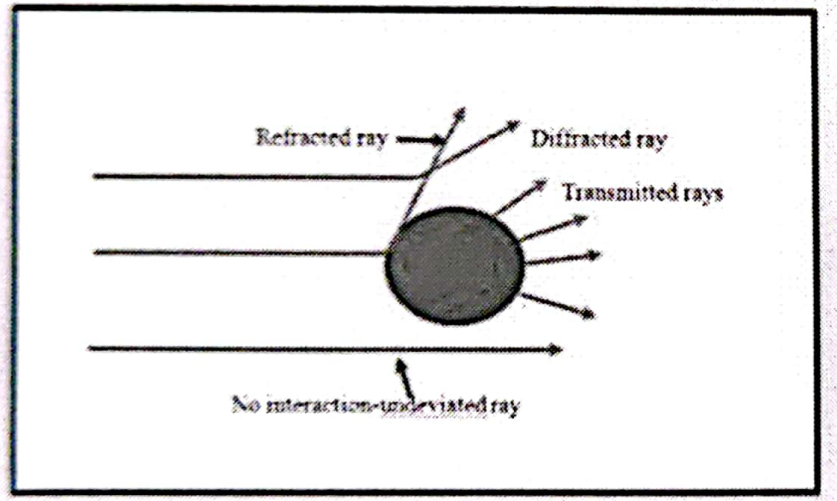


Figure: Modes of scattering of light in solution.

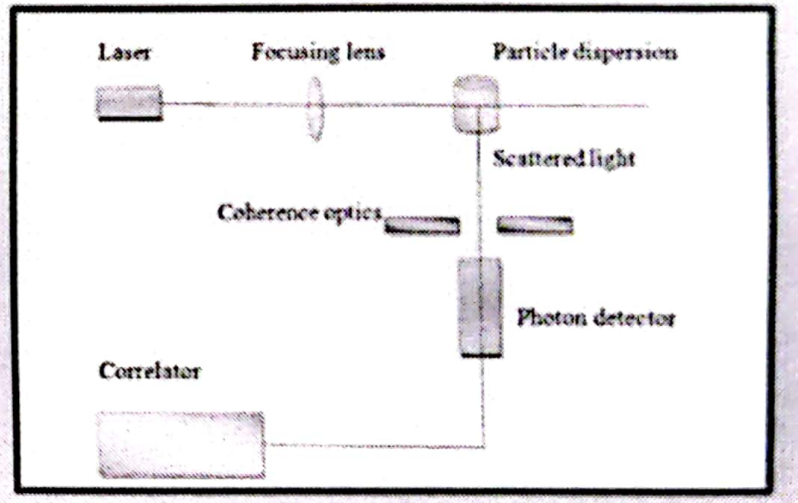
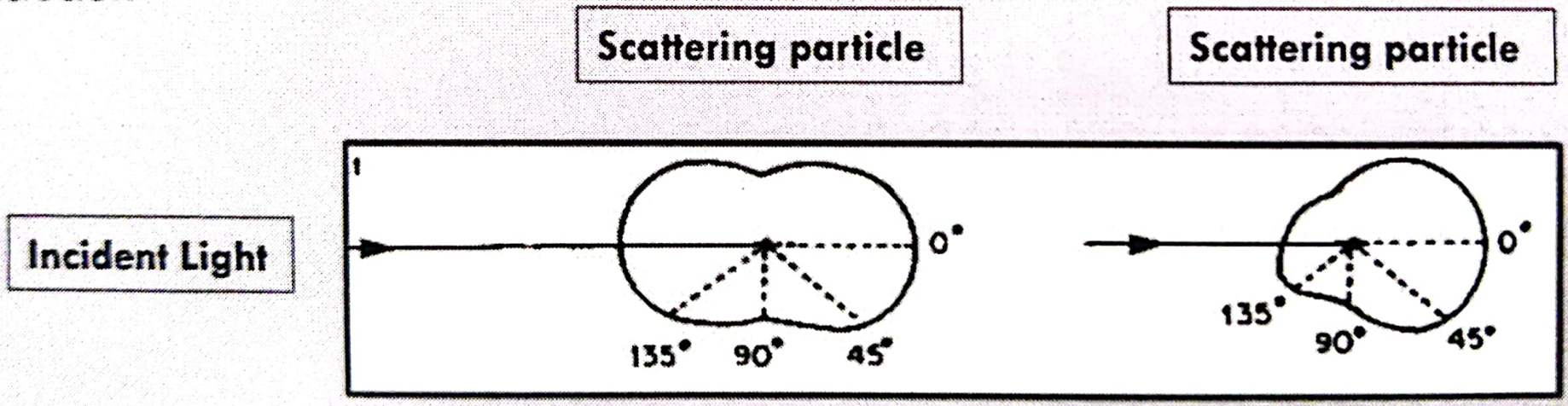


Figure: Schematic representation of light scattering.

For light scattering measurements, the total amount of the scattered light is deduced from the decrease in intensity of the incident beam, I0, as it passes through a polymer sample. This can be described in terms of Beer's law for the absorption of light.

In the determination of the weight-average molecular weight of polymer molecules in dust-free solutions, one measures the intensity of scattered light from a mercury arc lamp or laser at different concentrations and at different angles (θ), typically 0, 90, 45, and 135. The incident light sends out a scattering envelope that has four equivalent quadrants. The ratio of scattering at 45° compared with that for 135° is called the dissymmetry factor or dissymmetry ratio Z. The reduced dissymmetry factor Z0 is the intercept of the plot of Z as a function of concentration extrapolated to zero concentration



Light-scattering envelopes. Distance from the scattering particle to the boundaries of the envelope represents the magnitude of scattered light as a function of angle.

The Beer's law for the absorption of light as follows:

$$\frac{I}{I_0} = e^{-\tau l}$$

where is the measure of the decrease of the incident beam intensity per unit length l of a given solution and is called the turbidity.

The intensity of scattered light or turbidity (τ) is proportional to the square of the difference between the index of refraction (n) of the polymer solution and of the solvent n_0, to the molecular weight of the polymer (M), and to the inverse fourth power of the wavelength of light used (λ). Thus,

$$\frac{Hc}{\tau} = \frac{1}{M_w P_0}(1 + 2Bc + Cc^2 + \ldots)$$

where the expression for the constant H is

$$H = \frac{32\pi^2}{3}\frac{n_0^2(dn/dc)^2}{\lambda^4 N} \quad \text{and} \quad \tau = K'n^2\left(\frac{i}{}\right.$$

where n_0 index of refraction of the solvent, n index of refraction of the solution, c concentra the virial constants B, C, etc., are related to the interaction of the solvent, P is the particle scatte factor, and N is Avogadro's number. The expression dn/dc is the specific refractive increment a

- The weight average molecular weight value of scattering polymers in solution related to their light scattering properties that define by ,
- $KC/R(\theta) = Hc/\tilde{I} = 1/MW(P(\theta) \quad + 2A2C + 3A\ 3C2 + ...)$
- where K is the wave vector,.
- C is solution concentration,
- R(θ) is the reduced Rayleigh ratio,
- P(θ) the particle scattering function,
- θ is the scattering angle
- , A is the osmotic virial coefficients, where n_0 solvent refractive index,
- $K = 2\pi^2 n^2_0 (dn/dC)^2/Na\lambda 2$
-
-

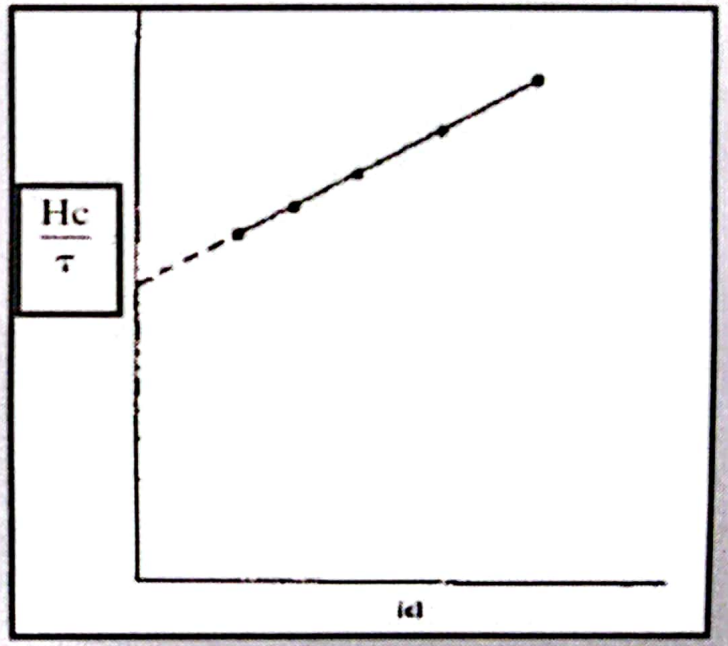
Several expressions are generally used in describing the relationship between values measured by light scattering photometry and molecular weight

$$Kc/R = 1/M_w (1 + 2Bc + Cc^2 + \ldots$$

At low concentrations of polymer in solution, the above equation reduces to an equation of a straight line (y = b + mx):

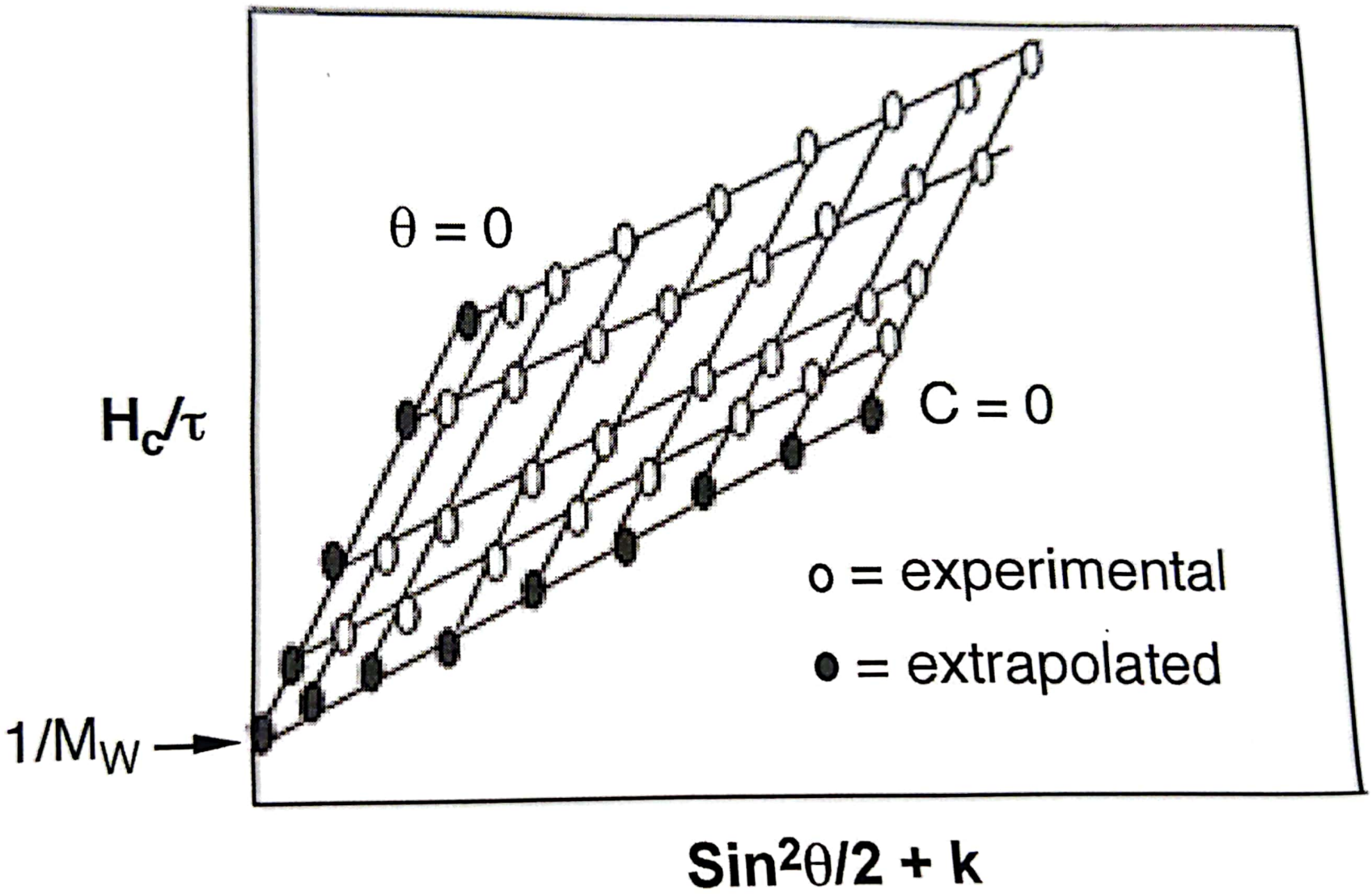
$$\frac{Hc}{\tau} = \frac{1}{\overline{M}_w} + \frac{2Bc}{\overline{M}_w}$$

When the ratio of the concentration c to the turbidity (related to the intensity of scattering at 0 and 90) multiplied by the constant H is plotted against concentration, the intercept of the extrapolated curve, is the reciprocal of Mw and the slope contains the virial constant B, as shown in Figure



Typical plot used to determine M_w^{-1}

where n_0 index of refraction of the solvent, n index of refraction of the solution, c concentration, the virial constants B, C, etc., are related to the interaction of the solvent, P is the particle scattering factor, and N is Avogadro's number. The expression dn/dc is the specific refractive increment and is determined by taking the slope of the refractive index readings as a function of polymer concentration.


θ = 0
C = 0
o = experimental
● = extrapolated
H_c/τ
$1/M_W$
$Sin^2\theta/2 + k$